

Work Measurement



- **Work study** is the systematic examination of the methods of carrying on activities so as to improve the effective use of resources and to set up standards of performance for the activities being carried out
- **Method study** the systematic recording and critical examination of existing and proposed ways of doing work as a means of developing and applying easier and more effective methods and reducing costs.
- **Work Measurement** is measuring time it should take a qualified worker to complete a specified task, working at a sustainable rate, using given methods, tools and equipment, raw material inputs, and workplace arrangement



MEANINGFUL STANDARDS HELP A FIRM TO DETERMINE-

1. Staffing needs(to meet required production)
2. Cost and time estimates prior to production
3. Crew size and work balance(who does what in an assembly line)
4. Expected production(to constitute a fair day's work)
5. Basis of wage-incentive plans(provides a reasonable incentive)
6. Efficiency of employees and supervision(standard is necessary to determine)



LABOR STANDARDS



Four ways of establishing labor standards:

1. Historical experience
2. Time studies
3. Predetermined time standards
4. Work sampling



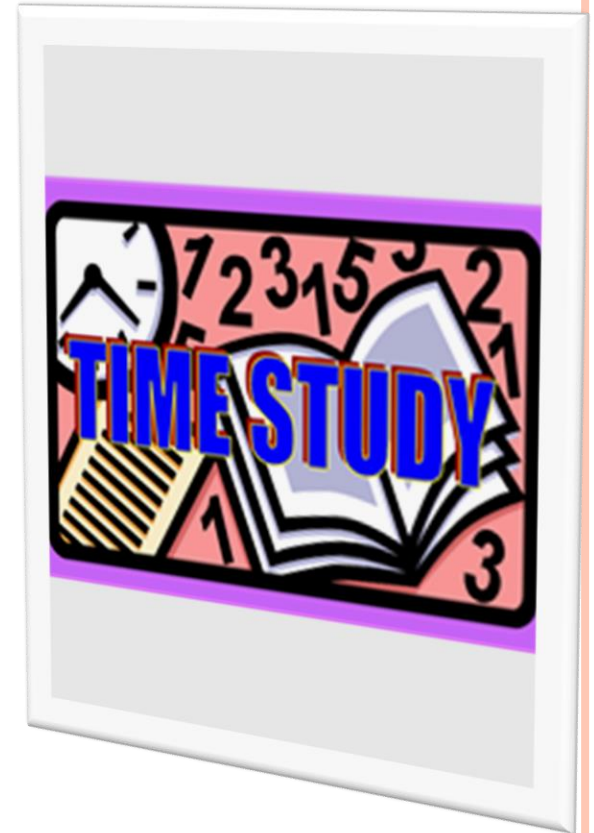
HISTORICAL EXPERIENCE

- ▶ How the task was performed last time
- ▶ Easy and inexpensive
- ▶ Data available from production records or time cards
- ▶ Data is not objective and may be inaccurate
- ▶ Not recommended



TIME STUDIES

- ▶ Involves timing a sample of a worker's performance and using it to set a standard
- ▶ Requires trained and experienced observers
- ▶ Cannot be set before the work is performed



TIME STUDIES

1. Define the task to be studied
2. Divide the task into precise elements
3. Decide how many times to measure the task
4. Time and record element times and rating of performance



TIME STUDIES

5. Compute average observed time

$$\text{Average observed time} = \frac{\text{Sum of the times recorded to perform each element}}{\text{Number of observations}}$$

6. Determine performance rating and normal time

$$\text{Normal time} = \text{Average observed time} \times \text{Performance rating factor}$$



TIME STUDIES

7. Add the normal times for each element to develop the total normal time for the task
8. Compute the standard time

$$\text{Standard time} = \frac{\text{Total normal time}}{1 - \text{Allowance factor}}$$



REST ALLOWANCES

- ▶ Personal time allowance
 - ▶ 4% - 7% of total time for use of restroom, water fountain, etc.
- ▶ Delay allowance
 - ▶ Based upon actual delays that occur
- ▶ Fatigue allowance
 - ▶ Based on human energy expenditure under various physical and environmental conditions



TIME STUDY EXAMPLE 1

- Average observed time = 4.0 minutes
- Worker rating = 85%
- Allowance factor = 13%

- Standard time??



TIME STUDY EXAMPLE 1

Average observed time = 4.0 minutes

Worker rating = 85%

Allowance factor = 13%

$$\begin{aligned}\text{Normal time} &= (\text{Average observed time}) \times (\text{Rating factor}) \\ &= (4.0)(.85) \\ &= 3.4 \text{ minutes}\end{aligned}$$

$$\begin{aligned}\text{Standard time} &= \frac{\text{Normal time}}{1 - \text{Allowance factor}} = \frac{3.4}{1 - .13} = \frac{3.4}{.87} \\ &= 3.9 \text{ minutes}\end{aligned}$$



TIME STUDY EXAMPLE 2

Allowance factor = 15%

JOB ELEMENT	OBSERVATIONS (MIN)					PERFORMANCE RATING
	1	2	3	4	5	
(A) Compose and type letter	8	10	9	21*	11	120%
(B) Type envelope address	2	3	2	1	3	105%
(C) Stuff, stamp, and seal envelopes	2	1	5*	2	1	110%

Calculate Standard Time



TIME STUDY EXAMPLE 2

Allowance factor = 15%

JOB ELEMENT	OBSERVATIONS (MIN)					PERFORMANCE RATING
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1. Delete unusual or nonrecurring observations (marked with *)
2. Compute average times for each element

Average time for A = $(8 + 10 + 9 + 11)/4 = 9.5$ minutes

Average time for B = $(2 + 3 + 2 + 1 + 3)/5 = 2.2$ minutes

Average time for C = $(2 + 1 + 2 + 1)/4 = 1.5$ minutes



TIME STUDY EXAMPLE 2

3. Compute the normal time for each element

Normal time = (Average observed time) x (Rating)

Normal time for A = $(9.5)(1.2) = 11.4$ minutes

Normal time for B = $(2.2)(1.05) = 2.31$ minutes

Normal time for C = $(1.5)(1.10) = 1.65$ minutes

4. Add the normal times to find the total normal time

Total normal time = $11.40 + 2.31 + 1.65$
= 15.36 minutes



5. Standard time for the job:

Standard time = Total normal time/(1 - Allowance factor)

$$= 15.36/(1-.15)$$

$$= 18.07 \text{ min}$$

Thus, 18.07 minutes is the time standard for this job.



PREDETERMINED TIME STANDARDS

- ▶ Divide manual work into small basic elements that have established times
- ▶ Can be done in a laboratory away from the actual production operation
- ▶ Can be set before the work is actually performed
- ▶ No performance ratings are necessary



Thank You

